THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The sum of the five consecutive numbers starting from 33 is: 1

(a) 165 **(b)** 180 **(c)** 175 **(d)** 170

2. Think of 12. Add 6, multiply by 4 and subtract 24. The result is: 1

(a) 48 **(b)** 42 **(c)** 54 **(d)** 16

3. The smallest digit that can replace * in 27*5 so that the number is divisible by 9 is: 1

(a) 8 **(b)** 1 **(c)** 5 **(d)** 4

4. If a number is divisible by both 4 and 3, it is also divisible by: 1

(a) 9 **(b)** 12 **(c)** 7 **(d)** 24

5. Is 7183 divisible by 11? 1

(a) Only if the last digit is even **(b)** Only if the number is a perfect square

(c) Yes **(d)** No

6. Is 7298 divisible by 6? 1

(a) No **(b)** Only if the number is a perfect square

(c) Yes **(d)** Only if the last digit is even

7. For the number 6184, (largest 4-digit arrangement of its digits) - (smallest arrangement) is: 1

(a) 7263 **(b)** 7083 **(c)** 7173 **(d)** 7182

8. The sum of the three consecutive numbers 48, 49 and 50 is: 1

(a) 145 **(b)** 144 **(c)** 147 **(d)** 150

9. The sum of the five consecutive numbers starting from 40 is: 1

(a) 210 **(b)** 200 **(c)** 215 **(d)** 205

10. Think of 15. Add 3, multiply by 5 and subtract 15. The result is: 1

(a) 78 **(b)** 75 **(c)** 20 **(d)** 72

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Take 17. Add 7, multiply by 2, then subtract 14. What is the result? How is it related to 17? 2

12. Use the divisibility rule for 11 to test 7316. 2

13. Apply one step of the Kaprekar routine to 4350. 2

14. Take 26. Add 3, multiply by 4, then subtract 12. What is the result? How is it related to 26? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Check whether 52624 is divisible by 8 and by 11. 3

16. Apply the Kaprekar routine to 8216 repeatedly until you reach 6174. Write each step. 3

17. Show that the sum of 38 and its reverse is divisible by 11. 3

OR

Check whether 80652 is divisible by 8 and by 11. 3

18. Apply the Kaprekar routine to 4390 repeatedly until you reach 6174. Write each step. 3

OR

Show that the sum of 74 and its reverse is divisible by 11. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Replace the stars so that the number 3*7* is divisible by both 10 and 3. Find all possibilities. 5

OR

Take 71 and its reverse 17. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

20. Replace the stars so that the number 7*6* is divisible by both 10 and 9. Find all possibilities. 5

OR

Take 85 and its reverse 58. Find the difference and show that it is divisible by 9. Prove it in general for $10a + b$. 5

END OF QUESTION PAPER — Total: 40 Marks

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